

TSIM-Tok

Visual State



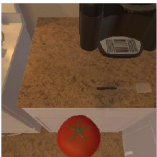
State 0



State 1



State 2



State 3

TSIM Router

Initial State

Fixed Token



V_0

TSIM (Medium)

Moderate Token



ΔV_1

TSIM (Low)

More Token



ΔV_2

TSIM (High)

Few Token



ΔV_3

Incremental Representation Module



Z_0

ΔZ_1

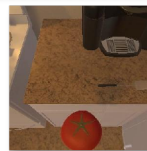
ΔZ_2

ΔZ_3

Incremental Visual Token

ViMo

S1: Go forward and face the counter to the right of the stove. <Image> \nS2: Pick the tomato up from the counter. <Image> \nS3: Turn left, go to the sink, turn the water on and off to wash the tomato, then pick it up <Image> \nS4: Go left and face the microwave above the stove. <Image> \nS5: Put the tomato in the microwave.



LLM



TSIM-Tok

Text Tokenizer



Please describe the sequence of actions the robot should take to transform the initial scene into the goal scene. The robot may need to prepare the tomato before placing it in its final location.

Und. Token

Visual Token

Text Token

Incremental Visual Token